

**Personal Rhythms of Memory: Investigating Individual Effects of Preferred Theta
Phase Lag between Sensory Inputs on Associative Memory**

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Abstract

Episodic memory formation requires integrating multisensory information. Inspired by animal research on how hippocampal theta oscillations modulate long-term potentiation, recent human studies have demonstrated that memory for auditory and visual stimuli modulated synchronously at theta frequency (4 Hz) is better than when those stimuli are modulated asynchronously; the so-called Theta-Induced Memory Enhancement (TIME) effect. However, recent failures to replicate the TIME effect question its robustness and dependence on other variables. One such variable may be inter-individual differences in neural sensitivity to the precise phase lag between auditory and visual streams. In this online study, we therefore tested a wider range of phase lags between 4 Hz modulated audio and visual stimuli. Using simulation methods from Zoefel et al. (2019), we estimated the sample size to achieve 90% statistical power to detect an optimal phase lag that varied across participants. However, this sample size showed no evidence that some phase lags produced better performance than others, thereby failing to replicate the TIME effect regardless of whether or not the optimal lag was assumed to vary across participants. Indeed, Bayes Factors supported the null for most analyses, and a follow-up analysis showed no evidence that the TIME effect depended on overall memory performance. Therefore, the present work adds to the growing number of non-replications of the TIME effect, though the reason for these failures remains unclear.

Introduction

Remembering a symphony performance that one attended involves retrieving the image of the orchestra, the music it produced, and the emotions that arose. Information from multiple sources must therefore have been integrated during the formation of that episodic memory. We know that long-term potentiation (LTP) in the hippocampus underlies such episodic memory formation (Kandel, 2001; Lynch, 2004), and electrophysiological research in rodents has observed facilitation or hindrance of LTP depending on whether the hippocampus is stimulated at the peak or trough of theta oscillations in hippocampal local field potentials, respectively (Hölscher et al., 1997; Huerta & Lisman, 1995; Hyman et al., 2003; Pavlides et al., 1988). Research with humans similarly demonstrated a link between theta oscillations in the medial temporal lobe (MTL) and episodic memory (Kota et al., 2020; Lega et al., 2012; Staudigl & Hanslmayr, 2013). These observations lead to the influential Separate Phases of Encoding and Retrieval (SPEAR) theory, which proposes that the phase of theta oscillations in the MTL modulates the formation and retrieval of episodic memories (see Hasselmo, 2025, for a review).

More recent research has raised the exciting possibility that theories like SPEAR can be tested non-invasively in humans. Biba et al. (2026) found that subsequent memory performance oscillated at ~ 7 Hz as a function of the onset of to-be-remembered objects relative to a phase-resetting task cue. They argue that this is a result of stimuli coinciding with different phases of endogenous theta oscillations. Though there is no evidence that the crucial phase resetting manipulation occurred as expected, the results support SPEAR. As it is difficult to track hippocampal theta oscillations in humans, other studies switched to entraining neural oscillations through exogenous modulation of stimuli at the theta frequency of 4-8 Hz (Haegens & Zion Golumbic, 2018; Wang et al., 2024). One can vary the synchrony

(phase-lag) between two such oscillating sensory streams (such as auditory and visual), to test whether memory for their association is affected. One might expect better integration into an episodic memory if the auditory and visual information from a symphony performance arrives in-phase at the hippocampus, at a frequency close to its endogenous theta rhythm.

This is indeed what was reported by Clouter et al. (2017) and Wang et al. (2018). They presented short (3s) clips in which the intensity of the visual and auditory streams was modulated at 4Hz, and varied the phase-lag between these two streams: either synchronous (0° phase lag) or asynchronous (90, 180, or 270°). They found that synchronous theta led to improved memory performance relative to asynchronous theta. They also found improved memory for synchronous theta relative to a non-modulated baseline condition, and relative to synchronous modulation at delta (1.7 Hz) or alpha (10.5 Hz) frequencies, suggesting that the advantage is specific to theta. They called this effect TIME (Theta-Induced Memory Enhancement). However, two recent studies by other research groups failed to replicate the TIME effect (Serin et al., 2024; Simeonov & Das, 2025). While there may be procedural differences that explain the non-replications, they at least warrant further investigation of the boundary conditions for the TIME effect.

One boundary condition may reflect individual differences in neural/behavioural sensitivity to the precise frequency of oscillations and/or precise phase-lags between oscillations. For example, Riecke et al. (2018) observed better performance in a speech perception task when participants received transcranial alternating current stimulation (tACS) at 4 Hz (the dominant amplitude modulation rate of their speech stimuli), but the optimal phase lag between the tACS and the speech signal varied across participants. Similarly, Cabral-Calderin et al. (2024) found that auditory detection performance was modulated by the phase

lag between tACS and the modulated audio signal, and the optimal phase lag varied between individuals.

Thus far, studies of the TIME effect have only tested for a single canonical phase lag for their synchronous condition. In fact, this “zero” phase lag tested previously was a value that was presumed to apply for the neural input into the hippocampus and allowed for slower transfer of visual information to the visual cortex than of auditory information to the auditory cortex. In previous work, therefore, the auditory stimulus was delayed by 40ms relative to the visual stimulus (Clouter et al., 2017; Wang et al., 2018).¹ In other words, the phase lag between the physical, 4Hz stimuli was actually $40/250 \times 360^\circ = 60^\circ$.

These previous studies, therefore, suggest both a neural mechanism and a way to test for individual differences in the optimal phase-lag needed for associative memory. Individual differences due to sensory processing speed or neural transmission time from auditory and visual sensory input to the hippocampus would result in different preferred phase relationships for encoding audio-visual episodic memories in single participants. One way to test for this is to more widely vary the phase-lag between the auditory and visual streams and to use an analysis approach described by Zoefel et al. (2019): fitting the resulting memory accuracy outcomes as a function of phase-lag using a (partial) Fourier set (see Statistical Analysis & Sample section for details). The goodness of fit of this Fourier set can be combined across participants to test for a consistent dependency of memory on phase-lag, even if the form of that dependency (e.g, optimal phase-lag) differs across participants. Another advantage of this approach is that it is also invariant to differences in stimulus

¹ Note this figure of 40ms, while derived from the neurophysiological literature, does not account for potentially different transmission times between auditory cortex and hippocampus, relative to visual cortex and hippocampus. Indeed, if the integration of auditory and visual information does require synchronous input to the hippocampus, humans should have evolved transmission delays that ensure this.

presentation across participants, e.g., consistent delays in presenting either audio or visual stimuli caused by hardware, internet browser, or other software differences. As long as audio and visual presentation latencies are stable over trials within each participant, we can run such an experiment online and still interpret results as indicative of the theta phase dependence of associative memory encoding.

We therefore ran an online experiment to test whether the phase-lag between 3-second clips of auditory and visual stimuli affects memory for the pairing of those stimuli. We used the same stimuli as Clouter et al. (2017), i.e., different types of instrumental music for the auditory clips, and extracts from documentaries for the visual clips, the pairing of which is arbitrary (i.e., not meaningful). After “encoding” blocks containing 16 of these pairs, memory for the pairs was tested by associative recognition, each trial of which replayed an auditory stimulus (unmodulated), and participants selected which of 4 visual stills came from the clip paired with that sound. The stimuli in the encoding blocks were always modulated at 4Hz, but by varying the phase-lag between auditory and visual streams across trials, we tested a range of conditions that include the synchronous and asynchronous conditions of Clouter et al. (2017).

If the TIME effect is replicable, we should see evidence that memory performance varies systematically as a function of phase-lag, even if the optimal phase-lag for memory differs across participants (see Table 1 for the complete study design table). If optimal phase-lag does differ substantially by participant, then this could help explain previous failed replications, in which the single phase-lag used may not have been sufficiently close to the optimal phase-lag for a sufficient number of the participants tested. Given the caveats offered above about how hardware or software differences can affect AV synchrony in online studies, we cannot determine with confidence if observed variation in optimal phase lag is entirely

due to participant variability. Nonetheless, our online method offers a more flexible test of the TIME effect compared to studies run in laboratory situations, which achieve consistent AV synchrony by using consistent hardware but assume negligible between-participant variation in optimal phase-lag. Alternatively, if there is no effect of phase-lag on memory performance in single participants in our online study, then the reliability of the TIME effect itself will be questioned.

Question	Does associative memory performance vary systematically as a function of the phase lag between auditory and visual stimuli at theta frequency?	Does the optimal phase lag for associative memory differ between individuals?
Hypothesis	Memory performance will be elevated around and peak at a certain phase lag compared to the rest of the phase lags.	The phase lag at which memory performance peaks will vary across participants rather than being fixed.
Sampling	384 young adults (18–35 years) will be recruited via Prolific. Sample size is based on simulations (Zoefel et al., 2019) to achieve 90% statistical power for detecting an effect of Cohen's $d = 0.4$.	Same sample.
Analysis Plan	We will use GLMs with sine and cosine predictors at 4 Hz to test whether memory accuracy varies sinusoidally with phase lag, and then combine participant-level p-values using Fisher's method. If non-significant, a one-sample Bayes test (Cauchy prior) will be conducted.	Reconstruct each participant's peak memory phase from the fitted functions. Use a Bayesian uniformity test (von Mises vs uniform distribution) to test whether preferred lags cluster or are uniformly distributed.
Rationale for deciding the sensitivity of the test for confirming or disconfirming the hypothesis	The effect size of Cohen's $d = 0.4$ for the power analysis was based on a meta-analysis of previous 0° vs 180° phase lag contrasts.	The study is powered to test the first and primary question.
Interpretation given different outcomes	Significant sinusoidal modulation would support the TIME effect. A Bayes factor ≥ 6 in favour of the null would be evidence against the TIME effect.	A Bayes factor ≥ 6 in favour of uniformity will be interpreted as an indication of the optimal phase differing across individuals. A Bayes factor ≥ 6 in favour of the von Mises distribution will provide evidence against the variance of optimal phase across individuals.
Theory that could be shown wrong by the outcomes	The Theta-Induced Memory Enhancement (TIME) effect.	That optimal phase lag between visual and auditory information varies across individuals.

Table 1. Study Design Table

Methods

Stimuli

We adapted the stimuli and the scripts for running the experiments from Clouter et al. (2017). The duration of each movie clip was 3 seconds. The video and sound pairs used to create these movie clips were the same as Serin et al. (2024), which randomly matched and checked these pairs to prevent apparent semantic connections. The videos were excerpts from documentaries involving natural or urban outdoor scenery. The sounds were divided into 8 categories and were orchestral or instrumental music with no recognisable lyrics. Each block had an equal number of sounds from four categories (e.g, acoustic, orchestra, choir, and synthesiser). These four sound stimulus categories changed every block, and their assignment to blocks was rotated across participants for counterbalancing.

MATLAB scripts were used to apply the sinusoidal modulation of visual and auditory intensities. The visual intensity started at 50% and then oscillated sinusoidally between 0 and 100% amplitude, while the auditory intensity started at a different initial amplitude (depending on the phase lag condition) and then similarly oscillated between 0 and 100% amplitude. For example, if a trial is in the 180-degree phase lag condition, the video brightness would start at 50%, reach 100%, go down to 0%, etc, while the audio volume would start at 50%, go down to 0% and then reach 100%. Importantly, we did not introduce any additional delay to the audio to account for the different processing times of visual and auditory information (i.e. we will exclude the 40 ms delay from Clouter et al., 2017), as we already cover a wide range of phase lags. Therefore, rather than testing the effect of some hypothesised neural lag, our phase lag conditions correspond to physical changes in the relative timing of sinusoidal modulations of sound and video playback. The experiment itself was programmed in JavaScript with the jsPsych toolbox (de Leeuw et al., 2023). For best

playback timing precision and reliability, video and audio files were combined with the H.264 and AAC codecs, respectively, and contained in .MP4 format to create a single file for each clip. The jsPsych “psychophysics” plugin (Kuroki, 2021) was used to play the clips.

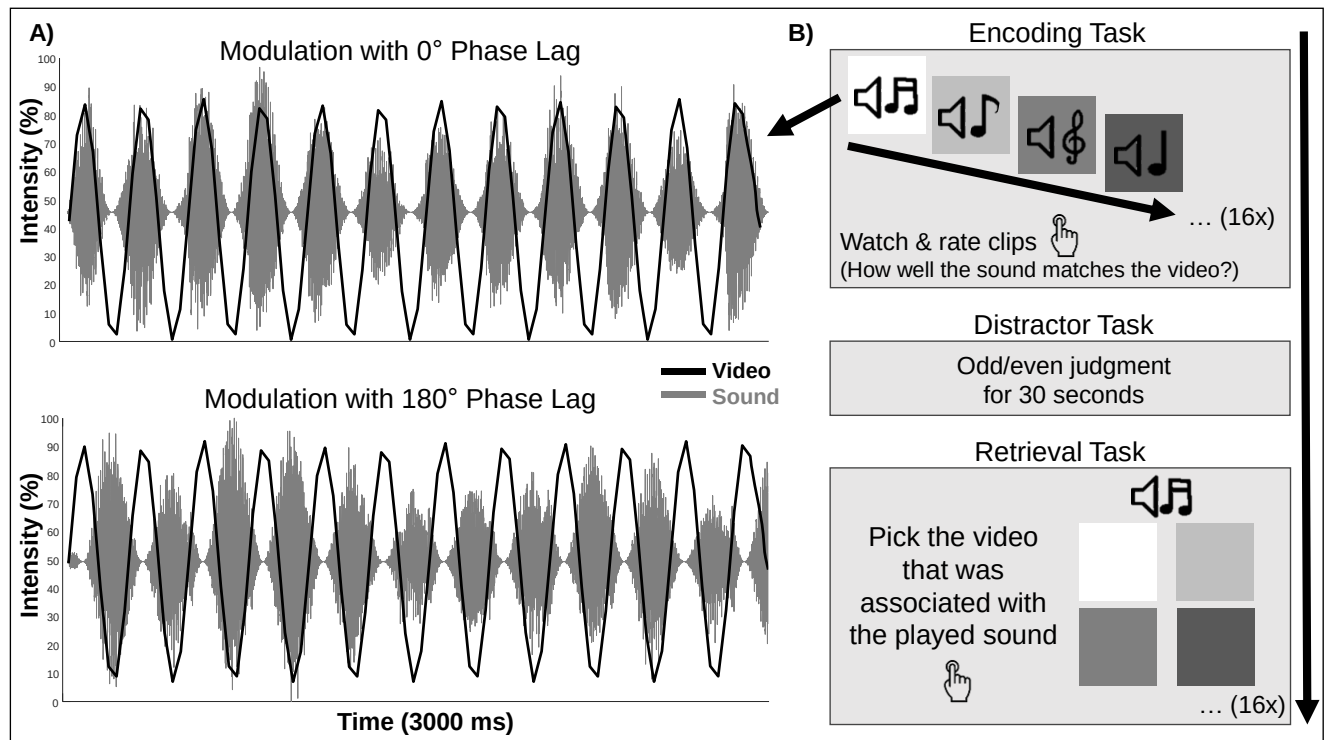


Figure 1. The modulation of the video and sound intensities is visualised with two examples of 0- and 180-degree phase lags in a sample clip (A). The video intensity was calculated as the average brightness among pixels in a frame. The right panel shows the experimental flow in an example block (B). Different musical notations and shades of grey symbolise different videos and sounds, respectively.

Procedure

The experiment consisted of 12 blocks, and in each block, participants watched 16 modulated movie clips and tried to remember them. The clips had different audio-visual phase lags depending on the condition. There were 16 phase-lag conditions, evenly spaced from -180 to +180 (i.e, phase-lag bins of 22.5°, or 15.6ms at 4Hz). Each block had one trial from each of

these 16 phase-lag conditions, but the order of these conditions was randomised within each block.

There were three Tasks per block. In Task 1, participants watched the 3000-ms movie clips (approximately² subtending a visual angle of 5.7°) on a black background, and heard the sounds through headphones, for which they were asked to set the volume to 50%. After each clip, they rated how well the sound and video matched, in order to encourage paying attention to both modalities (this was the memory encoding task used by all previous studies). A fixation cross followed each rating for an inter-stimulus interval, which was jittered between 1000 and 3000 ms. In Task 2, participants made odd-even judgments for random numbers between 1 and 99, for a period of 30 seconds, to reduce contributions from short-term memory and prevent rehearsal. In Task 3, participants' recall of the video-sound associations was assessed using a cued recognition test. There were 16 trials to test every clip presented in Task 1. In each trial, participants heard one of the sounds presented in Task 1 and tried to pick the associated video from four stills, one of them being the correct option and the others being lures. Lures were videos that were also shown in the same block (so the task cannot be solved by familiarity with the video itself) and were associated with the same sound category as the target clip (so the task cannot be solved by remembering the category of sound). The stills selected were the middle frame of the video.

The assignment of clips to each of 16 phase bins was counterbalanced by dividing the 192 clips into 16 sets of 12 clips, and rotating the assignment of sets to phase bins across participants. Therefore, a multiple of 16 participants was required for complete

² The size of the video frame was set using a standard laptop screen of 15.6 inches as it would be an intermediate size between smaller laptops and bigger desktop monitors. Therefore, the actual visual angle of the frame would vary across participants but, this should be negligible for the purposes of the visual flickering effects.

counterbalancing. The order of phase conditions within a block in Task 1 was randomised for each participant and block, as well as the order of testing in Task 3. All the scripts used to create the stimuli, condition orders, experiment, power analysis simulations, and analysis pipeline are available on this GitHub repository: <https://github.com/fserin/OptimalThetaPhase>

Statistical Analysis

We had two critical hypotheses: 1) Memory performance for each participant will be elevated around and peak at a certain phase lag compared to the rest of the phase lags, and 2) this optimal phase lag will vary between participants. Considering these expectations, we required a statistical approach that can test such a nonlinear effect while accounting for the random effect of participants. Zoefel et al. (2019) simulated different analysis approaches to compare their sensitivity to responses that have phasic characteristics on simulated data. They assumed the real effect is a sinusoidal function of different phase lags, showing a single peak and trough at a certain phase and its antiphase, respectively, but assumed a random preferred phase for each individual. They then showed best statistical inference (on the binary data produced each trial) occurred when using a single sine and cosine function (at the frequency of the modulation) as the predictors in a generalised (logistic) linear model. The p-values from each participant's fit can be combined to calculate a single p-value across participants using Fisher's method. In case the group p-value is not significant ($<.05$), we planned to conduct a one-sample Bayes test with a Cauchy prior to estimate evidence for the null, given that a Bayes test with a non-informative prior is proposed to be the same as Fisher's method (Kozioł & Tuckwell, 1999). A Bayes factor (BF) of 6 or higher in favour of the null hypothesis will be accepted as evidence for the null.

Assuming this first hypothesis is confirmed, to test our second hypothesis, we planned to reconstruct the peak of the fitted phase-function for each participant, and use a Bayesian

uniformity test (Mulder & Klugkist, 2021) to see whether these peaks were clumped by comparing uniform vs. a von Mises (circular normal) distribution. A BF of 6 or higher in favour of the uniform distribution would confirm our hypothesis that optimal phase lag varies across participants.

Sample

We conducted a power analysis using simulation methods adapted from Zoefel et al. (2019). To adapt these simulations to our requirements, we first modified the underlying effect to have no negative deflections, as we do not expect to find impaired performance for antiphase angle lag. This decision is based on previous findings that the out-of-phase conditions of 90, 180, and 270° do not differ in memory performance (Clouter et al., 2017). We simulated 192 trials and 16 equally-spaced phase bins of 22.5°, and a half-sine wave width of 90° for the positive deflection. The strength of the phase-lag preference corresponded to the area under this half-sine (henceforth the “area” parameter). To set this area parameter, we needed to simulate an effect size that matched previous data.

Based on a meta-analysis of the most tested comparison of 0- vs 180-degree theta phase lag, we found Cohen’s $d = 0.4$ (see <https://github.com/fserin/ReplicatingTIME> for the analysis). Given that Clouter et al. (2017) did find an effect when comparing 0 vs. 180-degree phase lags, we could not assume that the optimal phase lag for each participant is drawn randomly from a uniform distribution from -180 to +180°. Instead, we drew the optimal phase lag for each participant randomly from a triangular probability distribution that ranged between -45 and +45° of 0 (corresponding to the “synchronous” theta range used by Wang et al., 2018). Note that we still sampled phase lags (on each trial) uniformly across the whole range of -180 to +180°, to match what we planned for the main experiment below (in case the true

distribution of optimal phase lags is uniform). Based on these assumptions, a line search revealed that setting the area parameter to 5.5% corresponds to a Cohen's d of 0.4.

Having fixed the area parameter, we calculated the statistical power under a different assumption, namely that the optimal phase across participants is drawn from a uniform distribution between -180 and $+180^\circ$. Though this may seem unlikely (if people do have a preferred phase at all, then there is likely to be some cross-participant consistency in this), it seems the most conservative assumption for testing the possibility that failures to replicate the TIME effect in other studies are simply because they did not sample sufficient number of participants whose optimal phase lag was closer to 0° than 180° (or other categories of lag tested). In this scenario, Clouter et al. (and Wang et al.) were perhaps fortunate to preferentially sample sufficient numbers of participants whose optimal lag was close to 0. Note, however, that with the statistical tests used here, power to detect a theta phase effect is not influenced by whether preferred phases are uniformly distributed across participants, or grouped around zero or any other specific phase lag. Instead, statistical power to detect theta-modulation of memory encoding depends only on the area parameter, the number of trials of each phase lag, and the number of participants.

With these parameter values, we simulated 10000 iterations of different sample sizes in steps of 16 participants, and found that a sample size of 384 reached 90% statistical power for $\alpha = 0.02$, exceeding the conventional advice of at least 80% power. This sample size is a multiple of 16 and hence, suitable for counterbalanced assignment of clips to phase-bin conditions.

We recruited young adults aged 18 to 35 from the Prolific platform (www.prolific.com).

They provided informed consent and received monetary compensation. Participants had self-reported normal hearing, corrected-to-normal vision, and no history of photosensitivity, given

that the experiment involves rapidly flickering visual stimuli. Participants were excluded if their overall memory performance was at chance level (25%) according to a non-parametric permutation procedure. Excluded participants were replaced by recruiting new participants.

Prior to data collection, we conducted stimulus presentation latency tests on one Windows laptop (Dell Latitude 5410, Windows 11 - 24H2) and one Macbook (MacBook Pro 2017, MacOS 13.7.8) to ensure the relative timing of audio and visual presentation was sufficiently consistent over trials (ie. reliable within participant). According to our measurements on using Chrome, only the Macbook was able to show reliability below 5 ms (SD) across trials. Absolute differences in timing should, in theory, not affect our manipulation, because our design covers a whole cycle of phase lags and our statistical test is invariant to phase lag peak. Therefore, we decided to collect data from only the participants who used Macbooks and the Chrome browser to complete the experiment. Additionally, participants were asked to use wired headphones to further ensure stability of auditory presentation latency. Please see the supplementary materials (S1) for more details on these measurements.

We collected data from 456 participants on Prolific to reach our target sample size of 384 after exclusions. Using the aforementioned method to exclude participants who had near-chance-level overall accuracy, we excluded 67 participants. Furthermore, one participant admitted to not paying much attention throughout the task. Three other participants were accidentally assigned an already-used condition order, and hence removed from the analysis. Lastly, after an initial inspection of the responses, we observed that one participant was missing nearly 90% of responses during both the encoding and retrieval tasks. Therefore, in a minor deviation from our pre-registered exclusions, we introduced the criterion that participants should have at least half the responses present in each phase lag bin. Only one participant was excluded with this criterion. Following the exclusions, 384 participants

(Mean Age = 26.78, SD = 4.77, range = 18-35) were left from 456, which met the required sample size.

Results

Participants had an average accuracy of 39.7% (SD = 6.7%) across all phase lags (see Figure 2A) which is better-than-chance performance given the 4AFC memory task performed and as expected given exclusion of poorly-performing participants. For the first hypothesis of an effect of phase lag on memory accuracy, we ran the registered generalised linear model³ (GLM) on each participant's data separately:

$$Accuracy \sim \sin(\text{Phase Lag}) + \cos(\text{Phase Lag})$$

The combination of the coefficients was tested with a linear hypothesis test (Wald F-test) to produce a single p-value for the two predictors. These p-values were combined across participants using Fisher's Method (see Figure 2B for distribution of p-values). The outcome was not significant, $\chi^2(df = 768) = 778.94, p = .384$. Therefore, our results failed to support our first hypothesis.

Because the second hypothesis concerning consistent vs variable peaks across participants was tied to the confirmation of the first one, testing it became inconsequential.

Exploratory Analyses

A follow-up, unregistered analysis was used to estimate an effect size and a Bayes Factor for the null. We calculated each participant's effect size by taking the (square root) of the sum of

³ All statistical operations are performed in R Statistical Software (v4.4.2; R Core Team, 2024), using *poolr* package (Cinar & Viechtbauer, 2022) for Fisher's Method, *car* package (Fox & Weisberg, 2011), and *BayesFactor* package (Morey & Rouder, 2026) for all Bayesian approaches.

squared coefficients for the sine and cosine terms in the above model, i.e. $\sqrt{[(\beta_{\text{sine}})^2 + (\beta_{\text{cosine}})^2]}$. To correct for the positive bias inherent in this rectified measure, we subtracted a null effect size, which was estimated by averaging the corresponding effect size over 1000 random permutations of phase labels. The distribution of this corrected effect size across participants is shown in Figure 2C. The Bayes Factor for a one-sample T-test for the mean of these effect sizes ($M = 0.006$, $SD = 0.141$, Cohen's $d = 0.04$) being bigger than zero was $BF_{01} = 8.25$, i.e., favouring the null hypothesis. This result reinforces the claim that there was no effect of phase lag on memory accuracy.

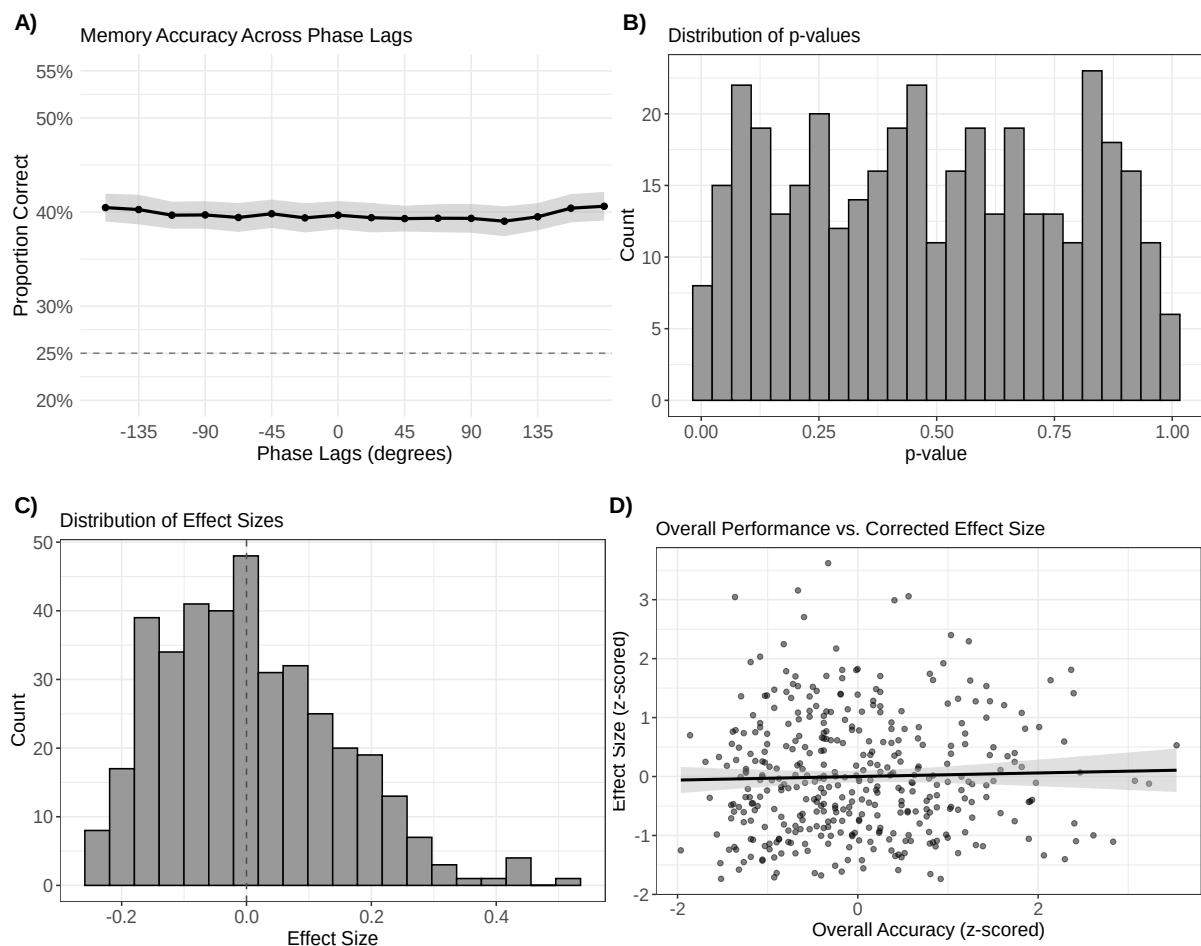


Figure 2. (A) Average accuracies are plotted across all phase lags. The shaded band covers the 95% confidence interval around the means and the bold line shows the grand averages. The dashed line shows the 25% correct responses. (B) The distribution of p-values from the

sine + cosine model per participant. **(C)** The distribution of corrected effect sizes (square root of sum of squares of sine and cosine coefficients) from the same model per participant. **(D)** The overall accuracies are plotted against the effect sizes.

We also explored whether there is an optimal phase lag that is consistent across participants, for which pooling the data across participants in a single model is likely to achieve higher statistical power than our registered approach above (which allowed for different optimal phases per participant). We fitted the same sine and cosine predictors to mean accuracy across trials for each phase lag using a Bayesian linear model with random intercept for participants. This model again favoured the null, $BF_{01} = 123.13$, suggesting that there was no consistent peak in accuracy across phase lags.

Thirdly, we tested whether the TIME effect is restricted to a tighter range of phase lags than could be fit using the two sinusoids in our pre-registered analysis (which had frequency $f_0=1/360^\circ$). In other words, by fitting expansions of phase lag using harmonics of 4 Hz, we wanted to see if the effect can be observed even if it appeared in fewer neighbouring phase lags than anticipated. We therefore added sines and cosines with frequencies f_0 , $2f_0$, $3f_0$ and $4f_0$ to the original registered model above. There was still no significant effect of phase lag, $\chi^2(768) = 639.04$, $p = 0.999$ (Fisher's method). Thus, there was no support for the possibility that the true peak was narrower (up to half-width of $(360/4)/2 = 45^\circ$).

Lastly, we wanted to address a similar concern from our previous test of the TIME effect (Serin et al., 2024). The original authors of the effect were concerned that our relatively low overall accuracy may not have left enough memory signal for the TIME effect to be detected. Therefore, we correlated the permutation-corrected effect sizes (defined in our first exploratory analysis above) with participants' overall accuracy scores. There was no significant relationship between overall memory performance and the effect of phase

modulation, $r = 0.03$, 95% CI $[-0.07, 0.13]$, $p = .56$ (see Figure 2D). To further investigate this, we also performed Fisher's Method only with the participants with high overall memory performance (based on the median split by overall memory accuracy, for whom mean accuracy was 45%, $SD = 5\%$). We repeated the one-sample Bayesian t-test on the effect sizes for the high performing participants as well. Fisher's Method showed no evidence for the effect of phase lag, $\chi^2(df = 384) = 383.43$, $p = 0.499$, while Bayes Factors again provided clear support for the null, $BF_{01} = 9.358$, $M = 0.003$, $SD = 0.135$.

Discussion

We aimed to detect the TIME effect regardless of the potential for individual differences in the stimulus timing that led to theta synchronicity between auditory and visual streams. According to our planned analysis, we did not observe any evidence that individual memory performance depended on phase lag. In fact, a follow-up Bayes approach favoured the null. Therefore, we conclude that our work failed to replicate the TIME effect (Clouter et al., 2017; Wang et al., 2018) despite using an approach that could account for different optimal phase lags across participants.

We performed several additional, exploratory analyses. First, we pooled data across participants in a single model, in case there was a consistent optimal phase across participants, given that this is likely to have higher statistical power. Again however, there was no evidence for an effect of phase lag on memory performance. We also added harmonics of the sine and cosine functions used to model phase lag, in order to capture narrower peaks, but still found no evidence for a preferred phase lag. Finally, we investigated whether any effect of phase lag depended on overall memory performance, with smaller or less apparent TIME effects for participants with lower memory performance. However, we

found no evidence that the TIME effect size correlated positively with overall memory performance.

Note that the original TIME effect was reported when using a phase lag of $\sim 60^\circ$ between the visual and auditory streams, on the assumption that the differential neural transmission times from visual and auditory cortices would result in a 0° phase lag in arrival at the hippocampus (Clouter et al., 2017). This was not actually one of the phase lags tested here (which were multiples of 22.5° , i.e. closest of 67.5°). However, it seems very unlikely that a difference of 7.5° (at 4Hz) would make a big difference to the results. Furthermore, subsequent researchers defined the range of -45° to $+45^\circ$ around 0° as synchronous, when labelling the phase lags extracted from the EEG signal (Wang et al., 2018). Therefore, we believe that small differences in exact phase lag cannot explain our failure to replicate the effect.

However, one limitation of our design is the lower number of trials per phase lag. Instead of the usual 48 trials (Clouter et al., 2017; Serin et al., 2024), here we had 12 trials per condition, due to testing 16 different phase lag bins in each participant. Our exploration of the fine-grained phase-range assured us that any optimal lag could be detected, and simulation work showed that this lower number of trials per phase does not reduce the sensitivity of statistical methods applied here if the total number of trials remains unchanged (Zoefel et al., 2019). Nevertheless, it is possible that this choice made it difficult to detect a very narrow TIME effect peak focused on a single phase lag across participants.

Another difference from previous studies is that the experiment was conducted online, recruiting only MacBook users. This decision was based on our in-house calculations (See S1 for details) to ensure high inter-trial reliability of audio-visual presentation timing and to avoid phase lag achieved deviating from intended timing on some trials. We therefore have reason to believe that timing reliability should be comparable with previous lab-based

experiments. Nonetheless, with remote testing, there is less control over the exact hardware and software being used, and less control of participant compliance. It therefore remains possible that data quality was not as high as lab-based studies. Conversely, the benefit of remote testing is, of course, that we could more easily achieve a sample size approximately ten times larger than that achieved in previous lab-based studies with a necessary increase of statistical power.

The current lack of a TIME effect is consistent with some previous non-replications, both within (Wang et al., 2026) and outside the original research group (Serin et al., 2024; Simeonov & Das, 2025). Wang et al. (2026), however, observed that the pre-stimulus alpha power extracted with MEG correlates with the TIME effect, and hence, proposed that the brain state prior to the stimulation might explain the mixed evidence in the literature regarding the effect. In their study, the canonical synchronous vs asynchronous (180°) theta comparison favoured the null, but the pre-stimulus alpha power in the relevant sensory cortices predicted theta phase entrainment (defined as proximity of the extracted phase lag to the intended phase lag of presentation), which, in turn, predicted subsequent memory performance. They conclude that this accounts for the non-replications in the literature. However, it remains unclear why this affected some studies but not others. Thus, we believe that further research is needed to establish the boundary conditions of the TIME effect. In the absence of this evidence we suggest either that the TIME effect is non-existent, or much smaller than previously claimed and hence of limited practical utility.

Conclusion

The benefit of synchronous modulation of visual and auditory information at Theta frequency on episodic memory - the TIME effect - was proposed both as a theoretical mechanism to explain the binding of sensory information, and as a practical method to improve memory

outcomes generally (Clouter et al., 2017; Wang et al., 2018). Yet, some studies have failed to replicate the effect, both within (Wang et al., 2026) and outside the original research group (Serin et al., 2024; Simeonov & Das, 2025). One reason for this could be individual variation in the optimal phase-lag between sensory streams. However, when allowing for such individual differences across the full range of phase lags, we again fail to replicate the TIME effect. Thus the boundary conditions for, and existence of, the TIME effect remain unclear.

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All the scripts used to create the stimuli, condition sheets, experiment, power analysis simulations, and the analysis pipeline are available on a GitHub repository (<https://github.com/fserin/OptimalThetaPhase>), as well as the manuscript, all the data, materials, and registrations except for the videos used as stimuli due to copyright restrictions.

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